Gabriel MARIADASS | Paris, France

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SUMMARY

M.Sc. MVA at ENS Paris-Saclay and M.Eng. at École Nationale des Ponts et Chaussées, with applied ML experience at Dassault Systèmes and Banque de France, ML research at LPSM and NTU, and portfolio projects such as a RAG system with hybrid retrieval, reranking, LoRA and an agent. Seeking an AI Engineer role on LLM and generative AI applications in Paris. Available immediately.

EDUCATION

ENS Paris-Saclay, France *2025 - 2026*
M.Sc. Mathematics, Vision, and Machine Learning (MVA)

- **Academic average**: 17.19/20 (with Highest Honors)
- **Key courses**: ALTEGRAD (text and graph learning, LLM fine-tuning with LoRA, embeddings, retrieval and reranking), Deep Learning, Bayesian ML, Computational Statistics, Graphs in ML, Time Series ML, Statistical Learning with Extreme Values

École Nationale des Ponts et Chaussées, France *2022 - 2026*
M.Eng. - Mathematical and Computer Engineering (IMI)
Preparatory classes at Lycée Henri-IV (CPGE PCSI/PC*, 2020 - 2022). Final year completed at ENS Paris-Saclay (MVA, 2025 - 2026)

EXPERIENCES

LPSM, France & NTU, Singapore *04/2026 - 08/2026*
AI Research Intern
Supervisors: Badr-Eddine Chérief-Abdellatif (CNRS, LPSM) and Jérémie Houssineau (NTU)

- Extended uCBOpt (Singh et al., 2026), a possibilistic variational inference method, to estimate neural network uncertainty
- Improved calibration (ECE 0.025 -> 0.014) and beat a 64-sample IVON ensemble with a single forward pass

Dassault Systèmes, France *01/2025 - 07/2025*
AI Engineer Intern

- Optimized an attention-based GNN for retrosynthesis (DGL to PyG): up to 38% faster on CPU at comparable top-1
- Benchmarked CPU and GPU inference latency and tested dynamic and static quantization
- Evaluated Bayesian hyperparameter optimization (+1.5 points top-1, larger model) and refactored the code

Banque de France, France *09/2024 - 12/2024*
Data Scientist

- Built neural-network challengers with embeddings for categorical variables on 700K loans for an ECB audit
- Reached AUC up to 0.84 and up to +0.14 Gini over the bank's internal IRB models, depending on segment
- Analyzed default-rate time series (PCA, dynamic factor model) and PD margins, noting interpretability limits

PROJECTS

- reg-rag, regulatory RAG: hybrid retrieval and reranking (recall@5 0.988 vs 0.916 dense), LoRA query rewriter, search agent
- price-api, ML prediction API: FastAPI, MLflow, Docker, GitHub Actions and Cloud Run, 90% conformal prediction interval
- ALTEGRAD challenge, graph-to-text retrieval and reranking: Kaggle score 0.487 -> 0.661

TECHNICAL SKILLS

- **LLM / NLP**: Transformers, Hugging Face, RAG, LoRA fine-tuning, embeddings, retrieval, reranking, LLM agents, evaluation
- **ML & Deep Learning**: PyTorch, scikit-learn, XGBoost, GNNs (PyG, DGL), time series, calibration, quantization
- **ML Engineering**: Docker, FastAPI, CI/CD (GitHub Actions), GCP, MLflow, pytest, Git
- **Programming**: Python, SQL, Bash
- **Mathematics**: Probability & Statistics, Bayesian Learning, Convex Optimization, Extreme Values

LANGUAGES & INTERESTS

- **Languages**: French (native), English (C1), Spanish (B1)
- **Interests**: Muay Thai, Running, Weightlifting, Nutrition, AI Tech Watch, Tech Entrepreneurship, Reading, Travel